NISHA IYER

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PROFESSIONAL SUMMARY

I am an electronics and communication engineering student focused on embedded firmware, low-power sensing, and dependable device-to-cloud systems. I have worked with STM32 and ESP32 platforms, FreeRTOS task design, communication buses, sensor calibration, and automated hardware-in-the-loop testing. I enjoy debugging across boundaries, from electrical signals and timing constraints to firmware state machines and cloud telemetry. I am seeking embedded software, firmware, or IoT internships where I can contribute to products that need careful resource use and reliable field behavior.

EDUCATION

Vellore Institute of Technology - Bachelor of Technology - BTech, Electronics and Communication Engineering — Expected May 2027 | Chennai, Tamil Nadu

Relevant Coursework: Embedded Systems, Digital Signal Processing, Microcontrollers, Computer Architecture, Communication Systems, Control Systems, Data Structures, IoT Security

EXPERIENCE

Embedded Software Intern | Asterion Mobility Labs — May 2026 - July 2026 | Chennai, India

- Implemented FreeRTOS tasks for battery sensing, motor-temperature monitoring, diagnostics, and CAN message handling on an STM32-based controller.
- Reduced telemetry packet loss from 4.7% to 0.6% by redesigning interrupt-to-task queues and adding bounded retry logic.
- Created Python hardware-in-the-loop tests for 38 fault conditions, including sensor disconnects, stale values, bus congestion, and brownout recovery.
- Documented timing budgets and stack usage for six tasks, enabling safer priority and memory configuration before road testing.

Student Research Assistant | VIT Embedded Intelligence Lab — September 2025 - Present | Vellore, India

- Designed a low-power vibration-sensing node for predictive maintenance experiments using accelerometer sampling, local feature extraction, and LoRaWAN uplinks.
- Optimized duty cycling and fixed-point signal processing to extend estimated battery life from 11 weeks to 19 weeks.
- Collected labeled motor-fault datasets and evaluated threshold, random forest, and one-class anomaly approaches for edge deployment.

Firmware Trainee | CircuitSpring Academy — December 2024 - March 2025 | Remote

- Completed project-based training in GPIO, timers, ADC, PWM, UART, SPI, I2C, debouncing, bootloaders, and defensive C programming.
- Built reusable peripheral drivers and unit-tested protocol parsing on the host before flashing target hardware.

SELECTED PROJECTS

Edge Predictive Maintenance Node / *STM32, FreeRTOS, C, CMSIS-DSP, LoRaWAN* — October 2025 - Present

- Built a sensor node that samples three-axis vibration, computes RMS, kurtosis, crest factor, and band-energy features, and transmits compact health summaries.
- Implemented ring buffers and double-buffered DMA to maintain deterministic acquisition while communication tasks were active.
- Added remote configuration with versioned settings, CRC validation, and rollback to last-known-good parameters.

Low-Power Posture Wearable / *ESP32, C++, BLE, IMU, Flutter* — June 2025 - September 2025

- Developed a wearable prototype that classifies sustained posture states and sends configurable feedback over Bluetooth Low Energy.
- Reduced average current draw by 37% using event-driven sampling, connection interval tuning, and deep-sleep transitions.

ADDITIONAL EXPERIENCE HIGHLIGHTS

- Worked across Asterion Mobility Labs, VIT Embedded Intelligence Lab, and CircuitSpring Academy in roles requiring independent implementation, structured review, and clear status communication.
- Combined Embedded C, C++, FreeRTOS with database, cloud, testing, and version-control workflows to move projects from initial requirements through documented validation.
- Balanced performance and reliability goals by measuring latency, throughput, data quality, or user completion before and after technical changes.
- Prepared design notes, test plans, demos, and handoff documentation so classmates, researchers, and internship teammates could reproduce results and maintain the work.

PROJECTS

CAN Bus Diagnostic Analyzer / *C++, Qt, SocketCAN, Python* February 2025 - May 2025

- Created a desktop tool for filtering, decoding, replaying, and exporting CAN frames with message-rate and timeout visualization.
- Added DBC-based signal decoding and scripts that generated repeatable fault-injection traffic.

Smart Irrigation Controller / *ESP32, MQTT, C, Node-RED, InfluxDB* August 2024 - December 2024

- Automated zone watering using soil moisture, rainfall, reservoir level, and configurable quiet hours.
- Implemented local fallback rules so schedules continued during broker or Wi-Fi outages.

SELECTED TECHNICAL CONTRIBUTIONS

- Designed project architecture for Edge Predictive Maintenance Node and Low-Power Posture Wearable, separating interface, application, data, and infrastructure concerns to support iterative changes.
- Implemented validation and error-handling paths for expected inputs, malformed records, partial failures, retries, and recovery scenarios rather than testing only the happy path.
- Used automated unit, integration, or hardware/system tests to verify core behavior and added repeatable fixtures or seeded datasets for regression testing.
- Profiled or benchmarked critical workflows and used measured results to prioritize indexing, caching, batching, concurrency, memory, or communication improvements.
- Created deployment or build automation with version control and documented environment setup, configuration, observability, and rollback expectations.
- Presented technical tradeoffs to peers and nontechnical stakeholders, connected implementation choices to user or operational outcomes, and incorporated review feedback.

LEADERSHIP & ACTIVITIES

- VIT Robotics Club - firmware lead for a six-member autonomous rover team.
- IEEE Student Branch - conducted beginner workshops on microcontrollers, sensor interfaces, and debugging with logic analyzers.
- Women in Hardware mentorship circle - paired with first-year students for semester-long build projects.

TECHNICAL SKILLS

Programming: Embedded C, C++, Python, MATLAB, Bash

Platforms: STM32, ESP32, ARM Cortex-M, FreeRTOS, Arduino, Raspberry Pi

Interfaces: CAN, UART, SPI, I2C, BLE, MQTT, LoRaWAN

Tools: STM32CubeIDE, PlatformIO, Git, CMake, GDB, logic analyzer, oscilloscope

Concepts: Interrupts, DMA, fixed-point DSP, low-power design, bootloaders, HIL testing

ACADEMIC APPLICATIONS

- Applied Embedded Systems, Digital Signal Processing, and Microcontrollers concepts when selecting data structures, algorithms, interfaces, and performance constraints.
- Used ideas from Computer Architecture and Communication Systems to design storage models, data flows, communication boundaries, or experiment plans.
- Connected classroom work in Control Systems to maintainable code organization, testing strategy, documentation, and team review practices.
- Completed technical writing and presentation deliverables explaining assumptions, limitations, evaluation methods, and next-step recommendations.
- Maintained reproducible repositories with setup instructions, sample data, issue tracking, and versioned changes suitable for independent review.

CERTIFICATIONS

- Arm Education: Embedded Systems Essentials - 2026
- NPTEL Introduction to Internet of Things - 2025

HONORS & AWARDS

- VIT Research Seed Grant - Student Contributor, 2026
- IEEE Hardware Hackathon Runner-Up, 2025

ADDITIONAL DETAILS

Professional Interests: production-quality systems, measurable performance, careful testing, accessible documentation, and cross-functional delivery.

Profile Status: Synthetic testing profile; no real employment, credential, or identity claim is represented.

Links: github.com/nishaiyer-embedded | www.linkedin.com/in/nisha-iyer-embedded-sample | nishaiyer.example.com